Agenda

* Already Learnt - Summary
* To learn - Overview
  1. what is ML?
  2. MLE vs SDE
  3. ML Tasks
  4. Types of Learning

Made with Goodnotes

- Introduction to ML & NN → 1 Month
- ML Supervised learning → 1 Month
- ML unsupervised learning

Advanced → 1 Month

DAV 1 → Numpy, Pandas, matplotlib, seaborn

DS ML Libraries

DAV 2 →

Proba & Statistics

=

DAV 3

Fundamental

Co ordinate Geo & linear algebra → how to work with high dimention

Calculus & optimization → GD,

Numpy:

X & Y → dot product ⇒ np.dot(X, Y)

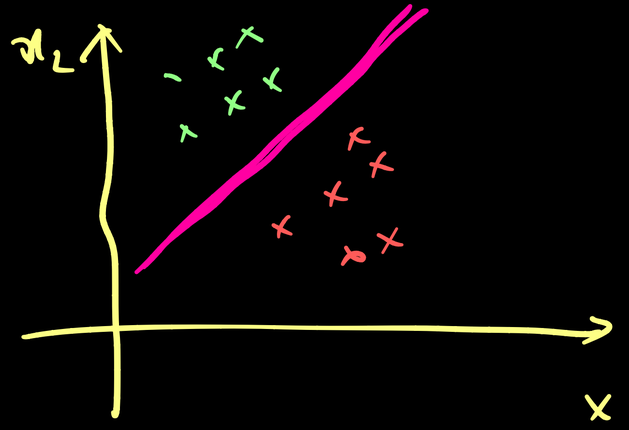
$\vec{w}$ & $\vec{x}$ → $\vec{w}^T \cdot \vec{x}$

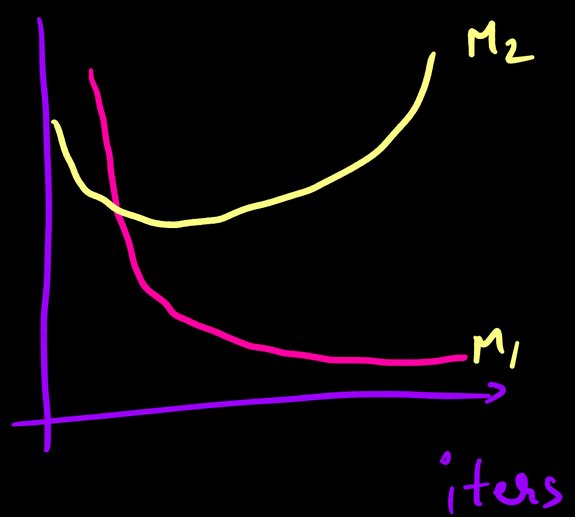
Pandas:

read & manipulating, null values

Matplotlib

seaborn





Made with Goodnotes

Univariate → 1 ✓

- Cat → bar, countplot, pie
- Num → histogram, KDE, Boxplot

Bivariate → ✓

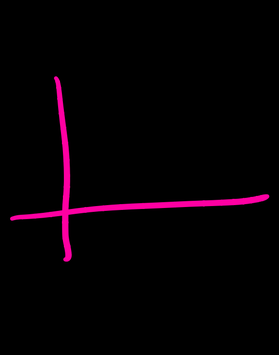
- Cat - Cat → dodged, stacked
- Cat - Num → Box plot, bar
- Num - Num → Scatter, line

You want to display the distribution of students' scores in a class. Which plot is best suited for this task?

0 users have participated

A	Bar Plot	0%
B	Histogram	0%
C	Scatter Plot	0%
D	Line Plot	0%

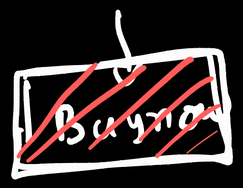
End Quiz Now



Made with Goodnotes

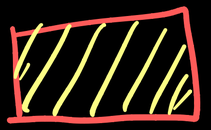
Prob & Stat

amazon



M1 ✓✓

Yes



M2

Yes

Conditional probability $P(\text{click} \mid \text{user}) = \overset{\checkmark\checkmark}{0.96}$

$P(\text{click} \mid \text{user}) = 0.88$

$\Rightarrow$ $z\text{-val} = \dfrac{x - \mu}{\sigma}$ (✓ over $x - \mu$)

$\Rightarrow$ $\mu = 0$ →

$\sigma = 1$

Made with Goodnotes

Coordinate / Lin algebra

$$y = mx + c$$

$$w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_0$$

$$\vec{w}^T \cdot \vec{x} + w_0$$

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Calculus & optimization

=> minimize the loss

=> GD => partial differenti

ML vs classical programing

Arthur Samuel :

ML provides ability to learn without being explicitly programed

without giving the logic

even (or) odd

```
n % 2 == 0
```

SDE

MLE

Made with Goodnotes

P.S Given an Email, you have to identify whether it is spam (or) not

How does the SDE pipeline look ?

1. look at the data, find some pattern → certain keywords

E_1 → S

E_2 → NS

E_3 → S

E_n → NS

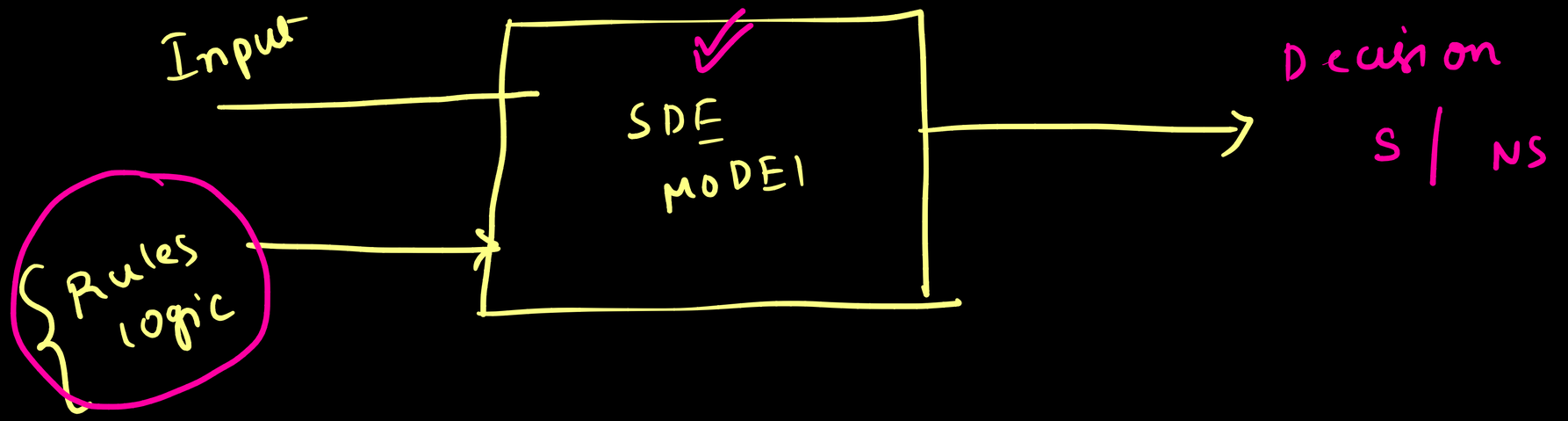
2. if - elif - else

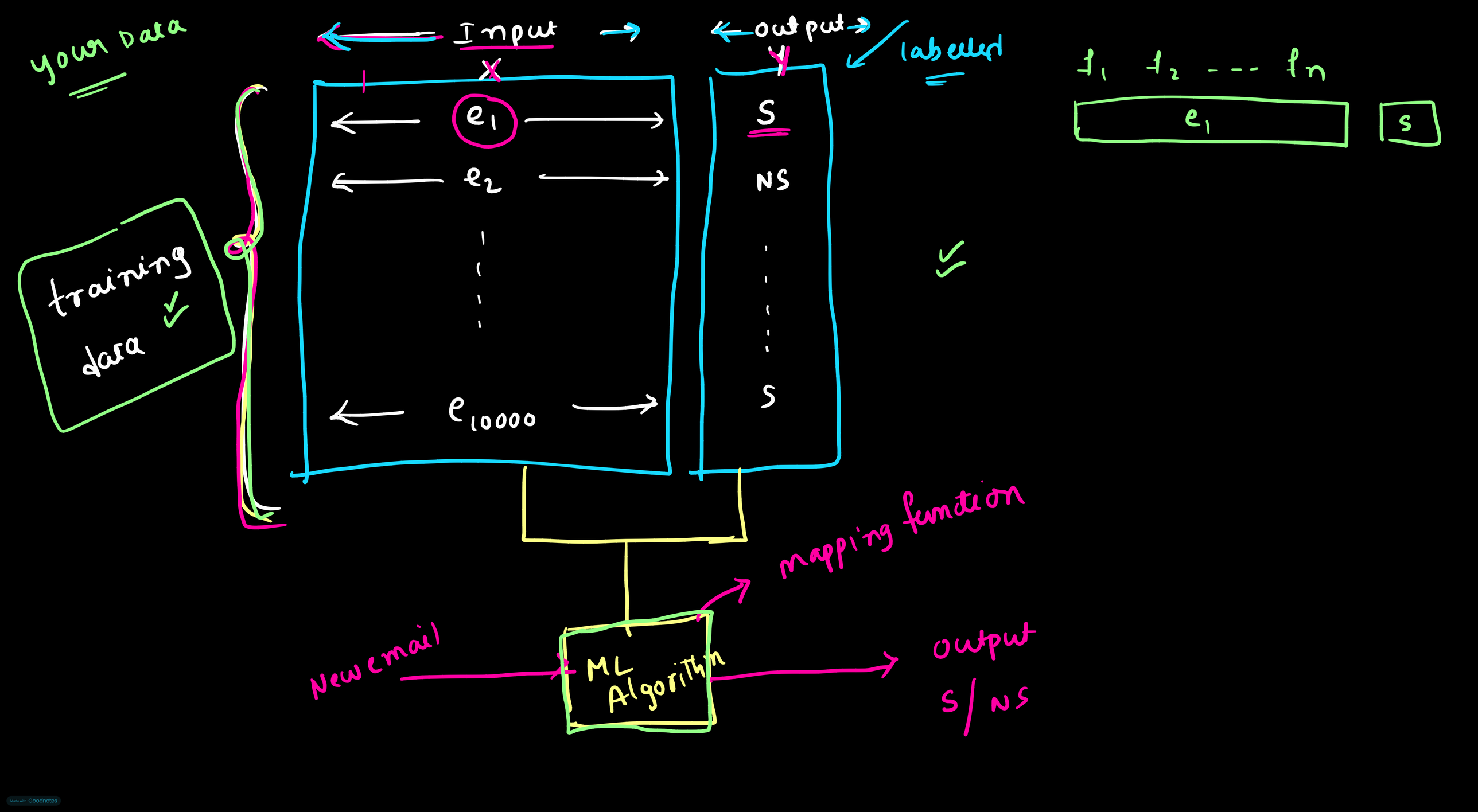
Keyword purchase / offer

if → S

NS

Made with Goodnotes


Input
SDE
MODEl
Decision
S / NS
Rules
logic


your Data
Input
X
output
Y
labelled
f_1 f_2 --- f_n
e_1
S
e_2
NS
e_{10000}
S
training data
mapping function
New email
ML Algorithm
output
S / NS

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.

E [Experience]

training data / historical data

→ past stock data

T [Task]

→ Spam | Not Spam

→ predict stock price

P [Performance]

accuracy

predicted vs actual

MSE → less

MAE

Made with Goodnotes

ML Topics categorization

Classification

→ Classify into one of the category

S/NS

→ spam / not spam

Regression

→ predict real value / continuous

→ House price prediction

Forecasting [Time]

Forecasting stock prices based on the last 7 days

Clustering

unsupervised

→ Group similar samples

Eg: customer segmentation

Recommendation

⇒ Netflix, Youtube

Made with Goodnotes

=> Supervised learning → output label is used for training

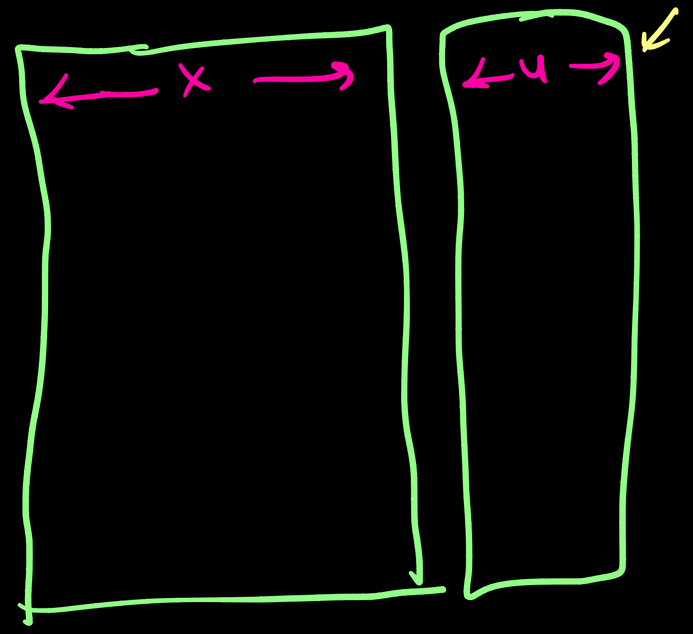
=> Unsupervised learning → output label not used for training

=> Reinforcement learning X

↳ Research labs

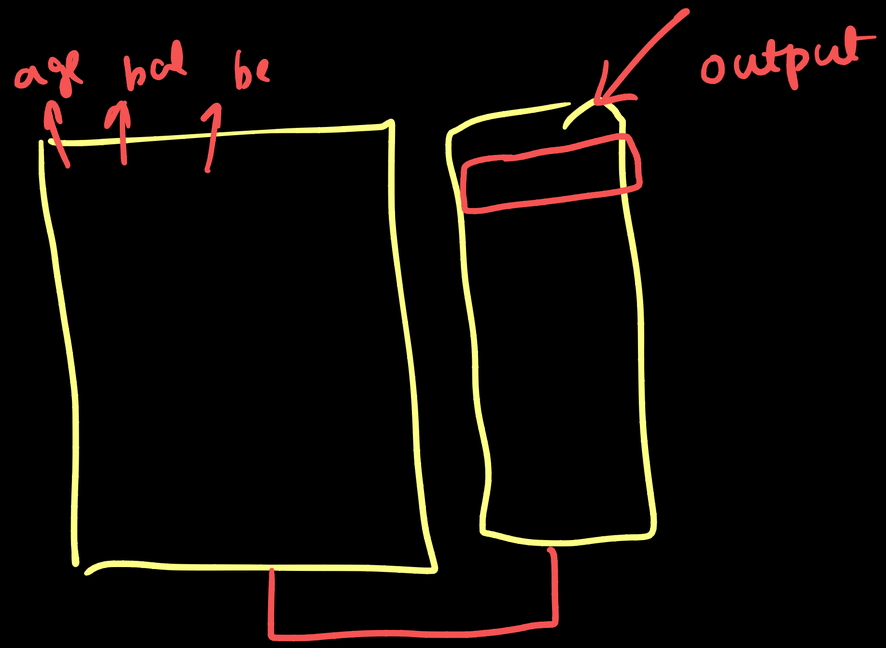
Made with Goodnotes

→ spam detection

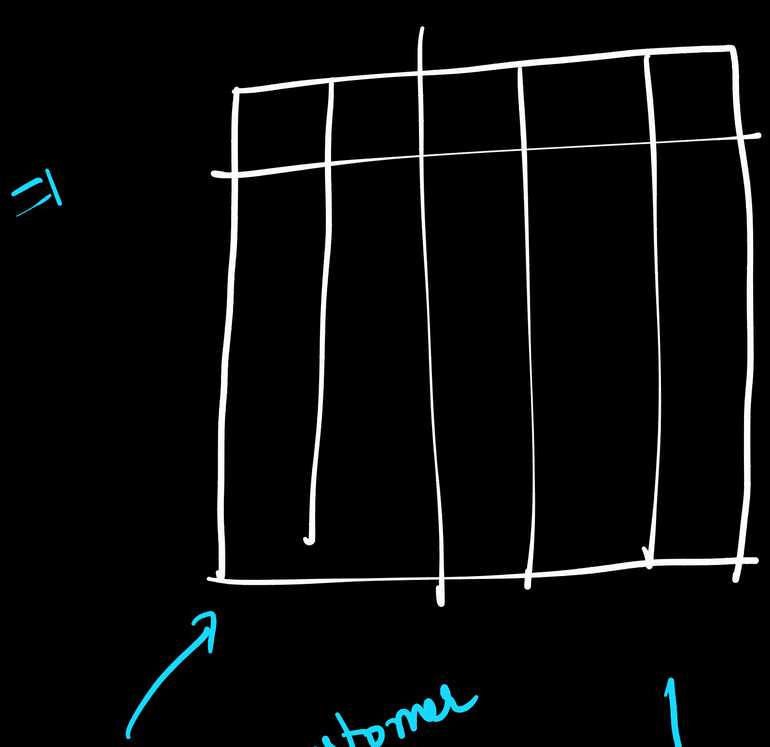


→ House price detection

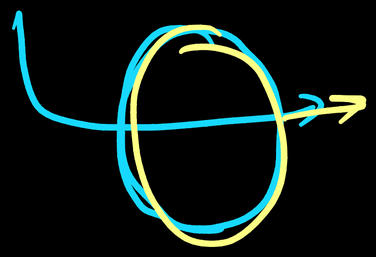
Supervised

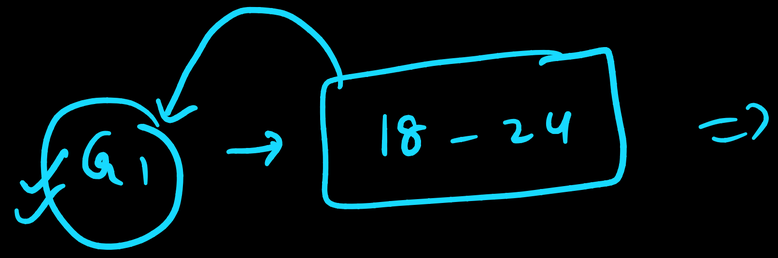


Made with Goodnotes



amazon customer data





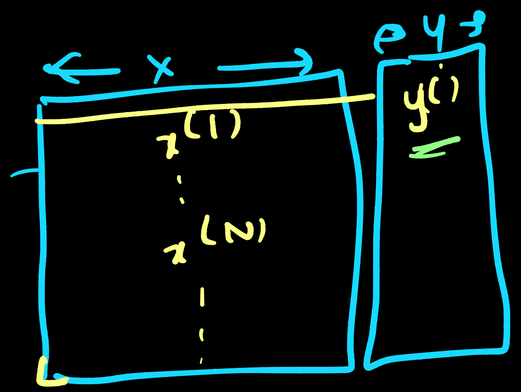
G2 → 25 - 35



Supervised learning

Dimension / feature / attribute

Classification



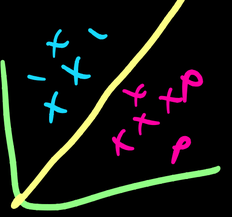
$$D = \{ (x^{(i)}, y^{(i)}) \}_{i=1}^{N}$$

$x^{(i)} \in \underline{R}^d$

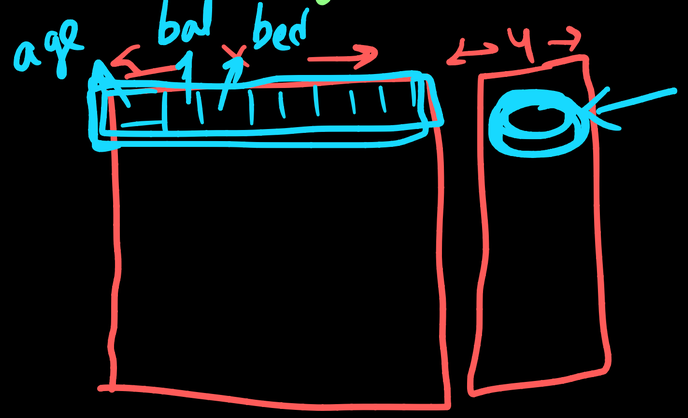
$y^{(i)} \in \{0, 1\}$

-1, +1

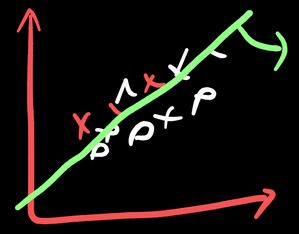
cat, dog



Regression



$$D = \{ (x^{(i)}, y^{(i)}) \}_{i=1}^{N} \; ; \; x^{(i)} \in R^d \; , \; y^{(i)} \in R$$

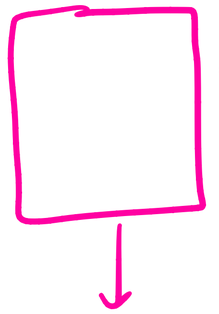


You have a dataset of customer feedback comments, and you want to categorize them into different topics, such as product quality, customer service, and delivery.
Is this a supervised or unsupervised learning problem?

0 users have participated

	A	Supervised learning	0%
✓	B	Unsupervised learning	0%

End Quiz Now



← unsupervised

7780568417

You are working on a project to predict stock prices based on historical market data, and you have a dataset with features such as past stock prices, trading volume, and other market indicators.
Is this a supervised or unsupervised learning problem?

0 users have participated

✓	A	Supervised learning	0%
	B	Unsupervised learning	0%

End Quiz Now

Made with Goodnotes